R Programming Lab 04

Image Classification Using a Neural Network

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1. Aim

To build and evaluate a neural network in R that classifies images into two categories using image preprocessing, numerical feature extraction, and supervised learning.

2. Tools and Dataset

The experiment uses the **jpeg** package to read images and **keras3** to build and train the neural network. The dataset contains 12 generated JPEG illustrations: six apples and six bananas. Each image is stored as an RGB image.

3. Image Exploration and Preprocessing

The notebook displays sample images and examines pixel-value summaries, a pixel histogram, and the image structure. Each image is resized from 128 × 128 pixels to 28 × 28 pixels. With three RGB channels, each resized image contains 2,352 numerical values.

4. Training and Testing Data

Subset	Images	Count
Training	apple1–apple4, banana1–banana4	8
Testing	apple5–apple6, banana5–banana6	4

Class 0 represents apples and class 1 represents bananas. The labels are converted to one-hot encoding so they match the two output units of the neural network.

5. Neural Network Architecture

Layer	Units	Activation
Input features	2,352	Pixel values
Hidden layer 1	256	ReLU
Hidden layer 2	128	ReLU
Output layer	2	Softmax

The model is compiled with the Adam optimizer, categorical cross-entropy loss, and accuracy as the metric. It is trained for 30 epochs with a batch size of 2, using 25% of the training data for validation.

6. Evaluation Results

Metric	Executed result
Test accuracy	100% (1.00)
Test loss	0.01148
Correct predictions	4 out of 4
Incorrect predictions	0

The model correctly classified all four test images. The apple images were predicted as Apple, and the banana images were predicted as Banana.

7. Confusion Matrix

Predicted \ Actual	Apple	Banana
Apple	2	0
Banana	0	2

The confusion matrix shows two correctly classified apples and two correctly classified bananas, with no misclassifications in this test set.

8. Conclusion

This experiment demonstrates an image-classification workflow using R and Keras 3. Images are loaded, resized, converted into numeric features, and labelled before being used to train a neural network. The model achieved 100% accuracy on the four test images. However, the dataset contains only 12 generated illustrations, so this result demonstrates the workflow rather than reliable real-world image recognition.